Roll No. | | | | | | | |

End Semester Examination 2025

Name of the Course: B.Tech. ECE (IOT) **Semester: V**
Name of the Paper: Transducers, Actuators and Display Devices **Paper Code: TEC 591**

Time: 3 Hours **Maximum Marks: 100**

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain the working principle and construction of a piezoelectric transducer.	**CO2/CO3**
(b)	Discuss the electrostrictive and magnetostrictive effects with neat diagrams.	
(c)	Explain the working of an electrostatic actuator with necessary equations. What are its advantages and limitations?	
Q2	(20marks)	
(a)	Describe the electromagnetic transducer. Explain how it can be used for sensing displacement.	**CO3**
(b)	Explain the working of an electrodynamic transducer. Discuss its applications in MEMS sensors.	
(c)	Describe the capacitive sensing with application.	
Q3	(20marks)	**CO4**
(a)	Explain the working principle of a PN homojunction solar cell with neat energy band diagrams.	
(b)	Describe the photovoltaic effect and explain how it is used in photovoltaic devices for solar energy conversion.	
(c)	Compare homojunction and heterojunction solar cells with suitable diagrams.	
Q4	(20marks)	
(a)	Explain the principle of electroluminescence. Describe the construction, working, and operation of an electroluminescent display with a neat diagram.	**CO5**
(b)	Explain the working principle of a plasma display with a neat sketch.	
(c)	Highlight the advantages and disadvantages of plasma displays compared to LCD and LED displays.	
Q5	(20marks)	
(a)	Explain the construction and working of a Liquid Crystal Display (LCD) with a neat diagram.	
(b)	Compare LCD vs LED displays in terms of technology, efficiency, and display quality.	**CO2/CO5**
(c)	Define the piezoresistive effect and derive the relation between resistance and stress.	